Progressive Compressively Sensed Band Processing for Hyperspectral Classification

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***Abstract*— Compressive sensing (CS) has recently been demonstrated as an enabling technology for hyperspectral sensing on remote and autonomous platforms. The power, on-board storage, and computation requirements associated with the high dimensionality of hyperspectral images (HSI) are still limiting factors for many applications. A recent work has exploited the benefits of CS to perform HSI classification directly in the compressively sensed band domain (CSBD). Since the required number of compressively sensed bands (CSBs) needed to achieve full band performance varies with the complexity of an image scene, this article presents a progressive band processing (PBP) approach, called progressive CSB classification (PCSBC), to adaptively determine an appropriate number of CSBs required to achieve full band performance, while also providing immediate feedback from progressions of class classification predictions carried out by PCSBC. By taking advantage of PBP, new progression metrics and stopping criteria are also designed for PCSBC. Four real-world HSIs are used to demonstrate the utility of PCSBC.**

***Index Terms*— Compressively sensed band progression confusion matrix (CSBPCM), compressively sensed bands (CSBs), compressive sensing (CS), hyperspectral classification, hyperspectral imaging, progressive compressively sensed band classification (PCSBC).**

List of Acronyms

AA	Average accuracy.
APR	Average precision.
CS	Compressive sensing.
CSB	Compressively sensed band.
CSBD	Compressively sensed band domain.
CSBPCM	Compressively sensed band progression confusion matrix.
CSBR	Compressively sensed band ratio.
HSI	Hyperspectral image.
OA	Overall accuracy.
OTC	Overall Tanimoto coefficient.
PBP	Progressive band processing.
PCSBC	Progressive compressively sensed band classification.
PGA	Progression accuracy.
PGAA	Progression average accuracy.
PGAPR	Progression average precision.
PGOA	Progression overall accuracy.
PGPR	Progression precision.
PR	Precision.
RIP	Restricted isometric property.
TC	Tanimoto coefficient.

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I. Introduction

HYPERSPECTRAL sensing technology has continued to find its success in a wide range of applications. Continued improvements in sensor design have also resulted in cheaper, smaller sensors that are now enabling new applications in remote and autonomous platforms. Unfortunately, many such platforms are still limited by the power, on-board storage, and computational requirements associated with the high dimensionality of HSIs. Contrary to multispectral images, HSIs have a large number of contiguous spectral bands which provide new information. However, given the finely sampled spectra, it is not surprising that hyperspectral pixels typically include a large amount of redundancy.

CS has been introduced as an enabling technology to facilitate data processing in the remote sensing community and for HSI in particular. CS approaches are based on directly sampling/sensing data in a compressed format. Such approaches are designed to remove redundancy from the data during the sampling process and can lead to savings in power requirements through reduced onboard computation, as well as reductions in on-board storage and required communication bandwidth. In addition, CS techniques can be applied to the spectral dimension, the spatial dimensions, or even all dimensions simultaneously, depending on the intended applications.

One of the current challenges with CS implementations in hyperspectral imaging is that traditional HSI processing algorithms cannot be applied to compressed data without additional considerations. Recent work [1] has exploited the benefits of CS to perform HSI classification directly on the data that is compressively sampled in the spectral domain. For convenience, the compressively sampled bands are referred to as CSBs and the resulting subspace is referred to as the

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CSBD. Given sufficient sensing conditions, [1] shows that full-band classifier performance could, indeed, be achieved in CSBD. Additionally, this work also highlighted the effects of scene complexity on determining the required number of CSBs to achieve acceptable classification performance. The experiments conducted in [1] showed that the optimal number of CSBs varied among sensors and image scene types. This observed variability motivated new research into how an appropriate number of CSBs can be selected.

This article investigates an adaptive estimation approach to addressing the above issue based on progressive observations as the number of CSBs changes. Its key idea is to look into PBP as an adaptive approach to determining the minimum required number of CSBs. PBP is a hyperspectral processing technique based on iteratively processing an image with full spatial information but reduced spectral bands [2]. PBP offers the benefits of providing immediate feedback, while the full image cube is being acquired and iteratively updating the input data cube as more spectral bands are processed. Using this approach, it is possible to transmit portions of HSIs in smaller, more efficient packets, while still generating near real-time feedback that improves performance as more spectral bands are processed.

Such a PBP framework is particularly suited for CSBD images. Specifically, we introduce a PCSBC that has two advantages over the traditional HSI classification approaches. First, the progressive nature provides immediate feedback without the need of collecting the full number of CSBs. As it was shown in [1], classification of some classes converged with a very small number of CSBs, which means that for some classes, a few progressions will already provide necessary informative results. Second, PCSBC determines the required number of CSBs directly from *in-situ* measurements that can work with any valid compressed classifier. By observing the behavior of such a band-compressed classifier in between progressions, a stopping criterion can be developed to adaptively determine when a sufficient number of CSBs have been acquired.

The proposed approach could potentially be implemented in two different applications. In the first application, additional sampling hardware can be added to a sensing platform to allow for compressed acquisition. An example might include a snapshot hyperspectral imager [3] that is used to generate instantaneous images at full spatial and spectral resolution, but that are never digitally sampled. Rather, digital sampling is limited to CSB mixtures that are collected serially. This approach would greatly benefit remote or unmanned platforms that require feedback from classification predictions to inform decisions or autonomy. Additionally, the classification progressions could be used to adaptively determine an appropriate number of CSBs allowing for more efficient use of the resources. As for a second application, the CSB-obtained HSIs can be fully acquired on the sensor platform, but the compressed analysis can be performed off-sensor on a control station. The CSBs can be sent to the control station serially and progressive band classification can be performed to determine when a sufficient number of CSBs have been received. This would allow for reduction in communication bandwidth and power consumption on the remote platform. This approach could be adapted to work with many different types of hyperspectral sensors.

Three major contributions are presented in this article. One is to develop a new approach, PCSBC. This is the first PBP algorithm ever proposed in the literature that operates in CSBD. Another is the introduction of a novel progression framework based on a new concept of CSBPCM. The CSBPCM-derived progression metrics are shown to correlate very highly with classification performance, making them well suited for designing stopping criteria. Furthermore, a connection is made between the proposed progression metrics and the well-known TC [4]. Finally, a fully automated stopping criterion is designed that is capable of adaptively determining an appropriate number of CSBs.

II. Compressed Classification

A. Universal Compressive Sensing Model

CS is a well-researched topic in the mathematical community and has received considerable attention over the past decade. It is a nonadaptive, sampling process that allows signals to be directly acquired in a compressed form and then later uncompressed using a sparse recovery process. CS techniques have potential to reduce hardware costs, on-board storage, computational costs, and communications bandwidths. While CS has predominantly been applied to spatial compression of multispectral images, its techniques can be readily extended to HSIs.

An HSI, $\mathbf{I} \in \Re^{N_x \times N_y \times L}$ consists of two spatial dimensions, N_x and N_y, and a spectral dimension L. A CSBD image, $\mathbf{I}_{\text{CSBD}} \in \Re^{N_x \times N_y \times m}$, can be formed by m CSBs, each of CSBs is a linear mixture of the original spectral bands. The compressed acquisition of a single hyperspectral pixel vector can be mathematically represented as follows:

$$\mathbf{r}_{\text{CSBD}} = \mathbf{\Phi}\mathbf{\Psi}\mathbf{r} + \mathbf{n} \tag{1}$$

where $\mathbf{r}_{\text{CSBD}} \in \Re^{m \times 1}$ is the compressed measurement vector, $\mathbf{\Phi} \in \Re^{m \times L}$ is the compressed sampling matrix, $\mathbf{\Psi} \in \Re^{L \times L}$ is the sparse representation basis, $\mathbf{r} \in \Re^{L \times 1}$ is a hyperspectral pixel vector, and $\mathbf{n} \in \Re^{m \times 1}$ is a noise term.

The model in (1) requires that the signals be sparse and sampled incoherently [5]. The challenge in designing a CS system is choosing $\mathbf{\Phi}$ and $\mathbf{\Psi}$ to satisfy these requirements for a given application. A previous work [6] showed that a hyperspectral model, well suited for analysis in the compressed domain, can be developed by using random sampling matrices. More specifically, $\mathbf{\Phi}$ is generated from any random distribution that satisfies the concentration of measure inequality [7]. Such random sampling matrices have been shown to be incoherent with any representation basis, $\mathbf{\Psi}$ and also satisfy the RIP that is often desired in compressed applications. RIP guarantees that relative measures of distance are preserved in the compressed space, effectively making the sampling matrix orthogonal $\mathbf{\Phi}\mathbf{\Phi}^T \approx \mathbf{I}$. Finally, when using a random sensing strategy, the sparse basis, $\mathbf{\Psi}$, only needs to be identified for sparse recovery. This universality property of random sensing also guarantees that all information is captured in the compressively sensed

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domain provided that a sufficient number of CSBs are used without knowing $\mathbf{\Psi}$. So, in this case, a simplified model can be derived from (1) as

$$\mathbf{r}_{\text{CSBD}} = \mathbf{\Phi_R}\mathbf{r} + \mathbf{n} \tag{2}$$

where $\mathbf{\Phi_R}$ represents a random sampling matrix.

B. Compressed Classification

For an HSI, $\mathbf{I}$, where each pixel belongs to one of M classes, a classifier is a function, $\delta(\mathbf{r})$, assigns a class label $j \in \{1, \ldots, J\}$ to each pixel vector $\mathbf{r}$. Based on the model shown in (2), it is possible to define a compressed classifier $\delta_{\text{CSBD}}(\mathbf{r})$ that operates directly on the CSBs

$$\delta_{\text{CSBD}}(\mathbf{r}) \equiv \delta(\mathbf{r}_{\text{CSBD}}). \tag{3}$$

In general, $\delta_{\text{CSBD}}(\mathbf{r})$ can be implemented using any classifier; however, there is no guarantee of equivalent classification performance in the compressed band domain. Furthermore, for a progressive band approach to be successful, the candidate-compressed classifier must provide equivalent or improved behavior as the number of CSBs is increased. For this reason, a compressed classifier must be chosen with care, and the expected behavior must first be analyzed and understood.

In this article, the compressed classifier presented in [1] is adopted and implemented exactly as described in the reference. This compressed classifier is adapted from the spectral-spatial classifier developed in [8] and is based on the support vector machine (SVM) [9] and a guided edge-preserving filter. The guided filter was applied to provide spatial context from the image, and since the image is not compressed spatially, no additional considerations are required for application in the compressed domain. The SVM, however, is directly affected by the spectral compression and was shown in [1] both analytically and empirically to converge to full band performance, given a compressive sampling matrix that satisfies the RIP. Additionally, the error between the compressed classifier and the full band classifier was shown to decrease as the number of CSBs is increased. This monotonic increase in classifier performance is an essential characteristic of a compressed classifier that is required to perform PBP in CSBD.

III. Progressive Compressively Sensed Band Processing for Classification

This section presents a new approach to PBP classification, called PCSBC, which consists of three key elements, progressive compressively sensed classifier, progression metrics, and stopping criteria. The proposed PCSBC must be capable of processing an HSI for varying numbers of CSBs in each progression. The progression metrics should be able to quantify the progressive changes in performance between successive progressions. Finally, the stopping criteria are definitive rules based on the progression metrics that determine when a number of CSBs should be sufficient. Each of them will be discussed in full detail in the following sections.

A. Progressive Compressively Sensed Classifier

To enable PCSBC to process data progressively, a desired classifier must be capable of processing images with a varying number of CSBs. To accomplish this, a general approach is developed based on employing a set of PCSBCs, $\{\delta^{m_i}(\mathbf{r})\}_{i=\min}^{\max}$ where is $\delta_{\text{CSBD}}(\mathbf{r})$ in (3) using m_i CSBs with m_i being the number of CSBs at the ith progression, $m_{i+1} = i + \Delta_i$, $m_1 = m_{\min} \geq 1$, $m_{\max} \leq L$, and Δ_i is a step size that adds new CSBs at the ith progression which can vary with each progression. Specifically, $\delta^{m_i}(\mathbf{r})$ is a classifier using m_i CSBs at the ith progression. However, in practice a larger step size and for most cases a lower maximum number CSBs will likely be sufficient.

To effectively train the set of classifiers, the compressive sampling matrix must be known *a priori*, and specifically included in the training process. Furthermore, the order of the columns in the sampling matrix must be preserved during the algorithm training and deployment. Both of these requirements can be easily satisfied by generating a random sampling matrix of maximum size, $\mathbf{\Phi} \in \Re^{m_{\max} \times L}$ and then selecting contiguous submatrices for each classifier. The training data are then projected into CSBD and the traditional classifier training procedure is used. This process is repeated until a desired set of classifiers has been trained. The complete training procedure is summarized in Algorithm 1. For simplicity, the training procedure has been described with uniform step size, i.e., $\Delta_i = \Delta$. Nevertheless, practically it is trivial to alter the training procedure to allow for a non-uniformly spaced number of CSBs.

Algorithm 1 PCSBC

Input: A hyperspectral training data $\mathbf{R}_N \in \Re^{N \times L}$ and sampling matrix $\mathbf{\Phi} \in \Re^{m_{\max} \times L}$ where there are NL-dimensional training samples to form a training matrix $\mathbf{R}_N$.

1. Choose an initial number of CSBs, $m = m_{\min}$ and a CSB step size $= \Delta$.
2. Form a sub-matrix, $\mathbf{\Phi} \in \Re^{m \times L}$ by selecting columns 1 to m from Φ and re-normalizing along the columns.
3. Project all of the training pixels into CSBD, $\mathbf{R}_{\text{CSBD}} = \mathbf{R}_N\mathbf{\Phi}$ with $\mathbf{R}_{\text{CSBD}} \in \Re^{m \times L}$.
4. Train the classifier, $\delta^{m_i}(\mathbf{r})$.
5. Increment the number of compressed bands, $m_{i+1} = m_i + \mathbf{\Delta}$.
6. Repeat steps 2 through 5 until $m = m_{\max}$.

Output: A set of trained classifiers, $\{\delta^{m_i}(\mathbf{r})\}_{i=\min}^{\max}$.

The presented approach has two desirable traits. First, this procedure does not require the classifier to be altered in any way. This allows for any valid classifier to be implemented without modification and for all of the standard classifier training procedures to be leveraged. Second, since PCSBC is not directly tied to the progression metrics or stopping criteria, the variations of PCSBC can be easily created by simply including a different set of classifiers. Moreover, it is even possible to train multiple sets of classifiers and employ classifier fusion techniques.

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The main shortcoming resulting from this approach is that for very complicated classifiers, such as deep neural networks, the required training time may become intractable. However, the free parameters, $\delta^{m_i}(\mathbf{r})$ and $m_{\max}$, provide a mechanism for trading between training time and progressive band resolution. Additionally, it is possible to develop adaptive training procedures that use the classification performance of the training data to update Δ_i to inform what $m_{\max}$ should be. Such adaptive training procedures are left for future work when more complicated classifiers are investigated.

B. Compressively Sensed Band Progression Confusion Matrix (CSBPCM)

Progression metrics are required to measure the change in classifier performance between progressive iterations. Such metrics are essential for determining if classifier performance has converged or if additional CSBs should be collected. As it was shown in [1], a compressed classifier performance tends to asymptote at a particular number of CSBs, based on the individual scene complexity. A progression metric is desired that will correlate well with the classification performance, since classification performance cannot be measured *in situ*. In this section, a new concept of CSBPCM is introduced, and a number of metrics are derived from it. A connection is made between the derived metrics and the well-known TC [4]. Finally, the progression metrics are cast into a probabilistic framework to provide additional insight into the subtleties of the proposed metrics.

Compressively Sensed Band Progression Confusion Matrix (CSBPCM): The proposed PCSBC leverages the specific nature of a classification problem to develop progression metrics that can be used for designing automatic stopping criteria. For each progression of PCSBC, a prediction of the class membership is made with each of the current data samples by a classifier δ^m using m CSBs.

Specifically, let $\hat{C}_j^{m_i}$ and $\hat{C}_j^{m_{i+1}}$ be the class memberships predictions by δ^{m_i} using m_i CSBs and $\delta^{m_{i+1}}$ using m_{i+1} CSBs, respectively, at two successive progressions, the ith and $i+1$th progressions where a "hat" is used to indicate that a class membership is predicted. The relationship between these two predicted class memberships can be exploited to provide several measures of progression.

Unlike the standard classification confusion matrix encountered in classification performance analysis where the ground truth is used against the classification, which is the predicted class membership, PCSBCM compares the class membership predictions between two progressions. More precisely, for each ith progression it replaces the ground truth with the M-class membership predictions, $\{\hat{C}_j^{m_i}\}_{j=1}^J$ obtained by δ^{m_i} using m_i CSBs at the ith progression against the J-class membership predictions, $\{\hat{C}_j^{m_{i+1}}\}_{j=1}^J$ obtained by $\delta^{m_{i+1}}$ using m_{i+1} CSBs at the $i+1$th progression. In terms of statistical estimation theory, $\{\hat{C}_j^{m_i}\}_{j=1}^J$ and $\{\hat{C}_j^{m_{i+1}}\}_{j=1}^J$ can be considered as *a priori* and *a posteriori* class membership predictions, respectively.

Assume that n_{ij} be the total number of data samples predicted by δ^{m_i} to be in the jth class at the ith progression but repredicted to be in the kth class by $\delta^{m_{i+1}}$ at the $i+1$th progression. Such interpretation treats the predictions for δ^{m_i} using m_i CSBs as the confusion ground truth and the predictions for $\delta^{m_{i+1}}$ using m_{i+1} CSBs as the confusion "prediction0."[1] It is motivated by the fact that predictions made using more CSBs will provide more accurate predictions, thus they should be assumed to be more reliable.

Let $\{\hat{C}_j^{m_i}\}_{j=1}^J$ be J classes predicted by δ^{m_i} using m_i CSBs.

1) $\mathbf{r}$ = data sample vector.
2) $\delta^{m_i}(\mathbf{r})$ = a classifier using m_i CSBs to predict the class membership of $\mathbf{r}$.
3) J = the number of classes of interest.
4) $\hat{C}_j^{m_i}$ = the set of data samples predicted by $\delta^{m_i}(\mathbf{r})$ to be in the jth class using m_i CSBs where a "hat" indicates a prediction.
5) $n_{ij}^{(m_i,m_{i+1})}$ = the number of data samples in the jth class, $\hat{C}_j^{m_i}$ to be classified into the ith class, $\hat{C}_i^{m_{i+1}}$.
6) $n_{jj}^{(m_i,m_{i+1})}$ number of data samples in the jth class, $\hat{C}_j^{m_i}$ correctly predicted to be in the jth class, $\hat{C}_j^{m_{i+1}}$.
7) $n_j = \sum_{i=1}^J n_{ij}^{(m_i,m_{i+1})}$.
8) N = total number of data samples, $N = \sum_{j=1}^J n_j$.

Fig. 1 shows a diagram of CSBPCM where for simplicity, n_{ij} is actually $n_{ij}^{(m_i,m_{i+1})}$.

Using CSBPCM and following an analogous approach to classification performance, it is possible to derive new progression classification criteria based on standard classification criteria, such as, OA, AA, PR, APR and overall precision (OPR), as follows:

$$\begin{aligned}\text{progression accuracy of } \hat{C}_j^{m_{(i,i+1)}} &= P_{\text{PGA}}\left(\hat{C}_j^{m_{(i,i+1)}}\right)\\ &= \frac{n_{jj}^{(m_i,m_{i+1})}}{\sum_{k=1}^J n_{kj}^{(m_i,m_{i+1})}} \qquad (4)\\ \text{progression precision of } \hat{C}_j^{m_{(i,i+1)}} &= P_{\text{PGPR}}\left(\hat{C}_j^{m_{(i,i+1)}}\right)\\ &= \frac{n_{jj}^{(m_i,m_{i+1})}}{\sum_{k=1}^J n_{jk}^{(m_i,m_{i+1})}} \qquad (5)\\ \text{progression overall accuracy} &= P_{\text{PGOA}}^{m_{(i,i+1)}}\\ &= \frac{n_{jj}^{(m_i,m_{i+1})}}{N} \qquad (6)\\ \text{progression average accuracy} &= P_{\text{PGAA}}^{m_{(i,i+1)}}\\ &= \frac{\sum_{j=1}^J P_{\text{PGA}}\left(\hat{C}_j^{m_{(i,i+1)}}\right)}{J} \qquad (7)\\ \text{progression average precision} &= P_{\text{PGAPR}}^{m_{(i,i+1)}}\\ &= \frac{\sum_{j=1}^J P_{\text{PGPR}}\left(\hat{C}_j^{m_{(i,i+1)}}\right)}{J}. \qquad (8)\end{aligned}$$

Relationship to the Tanimoto Coefficient: An interesting connection can be made between the proposed progression metrics and a binary similarity metric referred to as the TC.[2] In the context of a progression metric, TC, for the jth class $\hat{C}_j^{m_i}$

[1] It is important to note that the terms "prediction" and "truth" are used here to be consistent with the traditional classification confusion matrix; however, class membership ground truth is not available *in situ,* and this approach is, indeed, completely unsupervised.

[2] This is also referred to as the Jaccard Index (Jaccard 1901).

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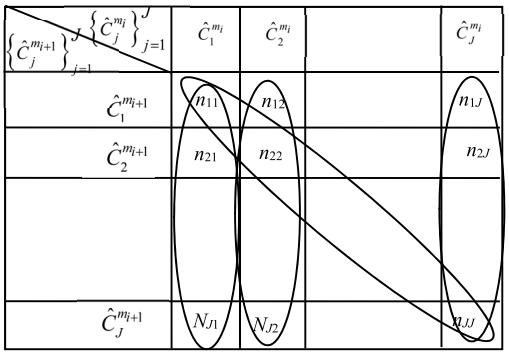


Fig. 1. Diagram of CSBPCM.

is defined as the ratio of the intersection of class predictions to the union of the class predictions

$$\mathrm{TC}_j^{(m_i,m_{i+1})} = \frac{|\hat{C}_j^{m_i} \cap \hat{C}_j^{m_{i+1}}|}{|\hat{C}_j^{m_i} \cup \hat{C}_j^{m_{i+1}}|} \tag{9}$$

where $|\cdot|$ denotes the number of elements in the set. In this article, $\hat{C}_j^{m_i}, \hat{C}_j^{m_{i+1}} \in \Re^{N_x \times N_y}$ is a Boolean matrix where a true value indicates that a particular data sample belongs to class $\hat{C}_j^{m_i}$. Also, $\mathrm{TC}_j^{(m_i,m_{i+1})}$ in (9) for a single class can be written in terms of the progression confusion matrix as

$$\mathrm{TC}_j^{(m_i,m_{i+1})} = \frac{n_{jj}^{(m_i,m_{i+1})}}{\sum_{k=1}^{J} n_{kj}^{(m_i,m_{i+1})} + \sum_{k=1}^{J} n_{jk}^{(m_i,m_{i+1})} - n_{jj}^{(m_i,m_{i+1})}}. \tag{10}$$

Moreover, (9) and (10) can be further explored by posing the problem in a probabilistic framework. Substituting in the basic definitions for a conditional and a joint probability, $P_{\mathrm{TC}(j)}^{(m_i,m_{i+1})}$, can be rewritten in terms of probabilities as

$$P_{\mathrm{TC}(j)}^{(m_i,m_{i+1})} = \frac{p\left(\hat{C}_j^{m_i} \cap \hat{C}_j^{m_{i+1}}\right)}{p\left(\hat{C}_j^{m_i} \cup \hat{C}_j^{m_{i+1}}\right)}. \tag{11}$$

Similarly, the proposed PGA and PGPR can also be rewritten in terms of probabilities, as shown in (12) and (13)

$$\begin{aligned} P_{\mathrm{PGA}}\left(\hat{C}_j^{m_{(i,i+1)}}\right) &= \frac{p\left(\hat{C}_j^{m_i}, \hat{C}_j^{m_{i+1}}\right)}{p\left(\hat{C}_j^{m_i}\right)} = \frac{p\left(\hat{C}_j^{m_{i+1}}|\hat{C}_j^{m_i}\right)p\left(\hat{C}_j^{m_i}\right)}{p\left(\hat{C}_j^{m_i}\right)} \\ &= p\left(\hat{C}_j^{m_{i+1}}|\hat{C}_j^{m_i}\right) \end{aligned} \tag{12}$$

$$\begin{aligned} P_{\mathrm{PGPR}}\left(\hat{C}_j^{m_{(i,i+1)}}\right) &= \frac{p\left(\hat{C}_j^{m_i}, \hat{C}_j^{m_{i+1}}\right)}{p\left(\hat{C}_j^{m_{i+1}}\right)} = \frac{p\left(\hat{C}_j^{m_i}|\hat{C}_j^{m_{i+1}}\right)p\left(\hat{C}_j^{m_{i+1}}\right)}{p\left(\hat{C}_j^{m_{i+1}}\right)} \\ &= p\left(\hat{C}_j^{m_i}|\hat{C}_j^{m_{i+1}}\right). \end{aligned} \tag{13}$$

Comparing (11) and (12)–(13), it is clear to see that $P_{\mathrm{TC}(j)}^{(m_i,m_{i+1})}$ is very similar to P_{PGA} and P_{PGPR}, but with a normalization term in the denominator that incorporates all predictions from both iterations, simultaneously. Another very important difference is that $P_{\mathrm{TC}(j)}^{(m_i,m_{i+1})}$ is a symmetric measure, while P_{PGA} and P_{PGPR} are not. More specifically, $P_{\mathrm{TC}(j)}^{(m_i,m_{i+1})}$ does not depend on the convention that is adopted to define the confusion matrix "truth," whereas, both P_{PGA} and P_{PGPR} depend on it.

Interestingly, for the adopted convention, the PGA is the *a priori* conditional probability and the PGPR is the *a posteriori* conditional probability. That is, P_{PGA} conditioned on the previous iteration, while P_{PGPR} is conditioned on the most recent iteration.

Finally, an OTC can be defined by summing up all classes simultaneously as

$$P_{\mathrm{TC}}^{(m_i,m_{i+1})} = \frac{\sum_{j=1}^{J} |\hat{C}_j^{m_i} \cap \hat{C}_j^{m_{i+1}}|}{\sum_{j=1}^{J} |\hat{C}_j^{m_i} \cup \hat{C}_j^{m_{i+1}}|}. \tag{14}$$

Comparatively, $P_{\mathrm{TC}}(m_i, m_{i+1})$ is the joint probability conditioned by the union of the marginal probabilities. It is again obvious from this form that $P_{\mathrm{TC}}(m_i, m_{i+1})$ is indeed symmetric and will not depend on the particular convention that is adopted.

By recognizing that the denominator of this expression is equal to the total number of data samples, N and that the intersection is the sum of the diagonal elements of the progression confusion matrix, $\sum_{j=1}^{J} n_{jj}$, it can then be shown that the overall PGA and the overall TI are indeed equivalent

$$P_{\mathrm{OTC}}^{(m_i,m_{i+1})} = \sum_{j=1}^{J} \mathrm{TC}_j^{(m_i,m_{i+1})} = \frac{\sum_{j=1}^{J} n_{jj}^{(m_i,m_{i+1})}}{N} = P_{\mathrm{OPGPR}}^{(m_i,m_{i+1})}. \tag{15}$$

Each of these metrics is explored in Section V and an analysis is performed to select the most appropriate measure for use with a stopping rule.

C. Stopping Criteria

Stopping criteria must be defined, based on the progression metrics, to indicate when enough CSBs have been collected. There are a few specific considerations that can be made for the progressive band convergence. First, all of the progression metrics defined in Section III-B are normalized between 0 and 1, which allow for target values to be directly included in a stopping rule. Second, in a noise-free system, the progression metrics should all be monotonically increasing, which can further reduce some of the complexity in the stopping rules.

To identify convergence in progression, a set of rules based on the absolute progression metric value and the magnitude of the difference in successive iterations are proposed. The stopping rule is based on two user inputs: a progression metric threshold τ_{PM} and a difference threshold τ_δ. The first threshold is simply the minimum progression metric that must be reached before convergence can be declared. This threshold is useful for preventing the algorithm converging at local minimum due to noise in the system. The second threshold represents the maximum change that is tolerated for a converged sequence. Values that are greater than this threshold are considered to have not yet converged. To make this threshold more intuitive, it is applied as a relative percentage of the current progression metric value. For any arbitrary progression metric using m_i CSBs at the ith progression, denoted by $\mathrm{PM}(m_i)$ and using m_{i+1} CSBs at the $i+1$th progression, denoted by $\mathrm{PM}(m_{i+1})$, the stopping rule can be defined as shown in Algorithm 2.

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Algorithm 2

If $\mathrm{PM}(m_{i+1}) - \mathrm{PM}(m_i) > 0$
  If $\mathrm{PM}(m_i) > \tau$ and $\frac{\mathrm{PM}(m_{i+1}) - \mathrm{PM}(m_i)}{\mathrm{PM}(m_i)} < \varepsilon$
    Declare convergence
  Else
    Continue
Else
  Ignore negative values

IV. Experiments

A. Experimental Setup

In order to conduct a comparative study with the work in [1], the same experimental procedure was performed to calculate CSBPCM where the same number of training/testing samples and classification hyperparameters were used. However, to properly simulate the *in situ* nature of the PBP, the results from a single trial are analyzed rather than the average of a collection of experiments.

For brevity, the performance metrics presented within this section are limited to P_{PGOA} and P_{PGAA}. Nevertheless, the results of P_{PGPR} were also in agreement. Additionally, the CSBR and performance efficacy metrics defined in [1] are both considered in our experiments. CSBR is simply defined by the ratio of the number of CSBs to the total number of spectral bands CSBD $= m/L$. Similarly, the performance efficacy $P_{\mathrm{EFF}} = P_{\mathrm{CSBD}}/P_{\mathrm{full}}$ is defined by the ratio of a performance metric measured in the CSBD to its corresponding performance metric measured using the full band image. These metrics allow us to have a consistent comparison by varying spectral bands.

Four real-world HSIs were used for our experiments. The first two images are the Purdue Indian Pines scene [10] as shown in Fig. 2 and the Salinas scene [11] as shown in Fig. 3. Both of these scenes were collected using the airborne visible/infrared imaging spectrometer (AVIRIS) [12] and contain 16 agricultural classes and a background class. The number of training and test samples used for the experiment is shown in Tables I and II, respectively.

The second two images are the University of Pavia scene [11] as shown in Fig. 4 and the Pavia Centre scene [11], as shown in Fig. 5. Both of these images were collected using the reflective optics system imaging spectrometer (ROSIS) [13] and contain nine urban classes and a background class. The number of training and test samples used for the experiment is shown in Tables III and IV, respectively.

B. Training Step Size and Maximum Bound

One potential concern is that for classifiers with a computationally burdensome inference process, using small step sizes of progression may become intractable. This thus raises the question of how fine the step size must be to adequately capture the performance progression. To answer this question, a simple experiment was conducted and only P_{OA} was calculated. The step size on the classification progression. The spectral–spatial classifier was trained on the Purdue Indian Pines image with a uniform step size of $\Delta_i = 1$ at each iteration. Multiple subsets of results were then created by subsampling with different uniform step-sizes $\Delta_i = 2, 5, 7$, and

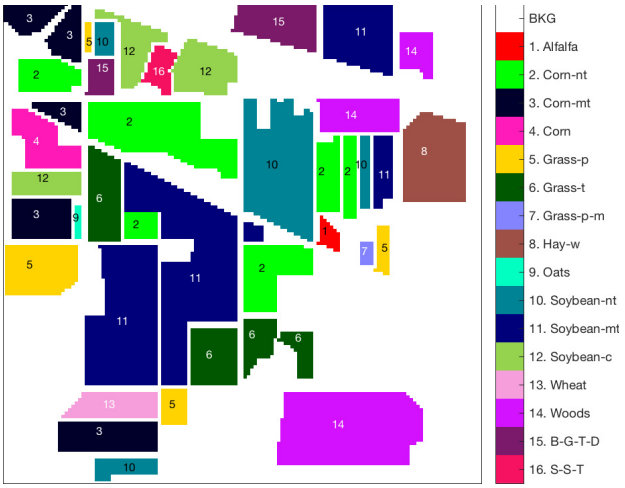


Fig. 2. Indian Pines along with ground truth image.

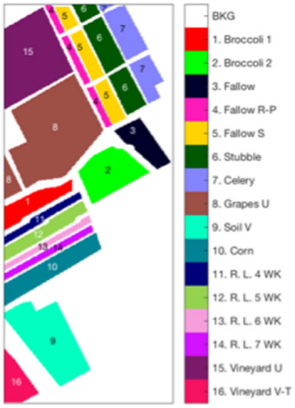


Fig. 3. Salinas along with ground truth image.

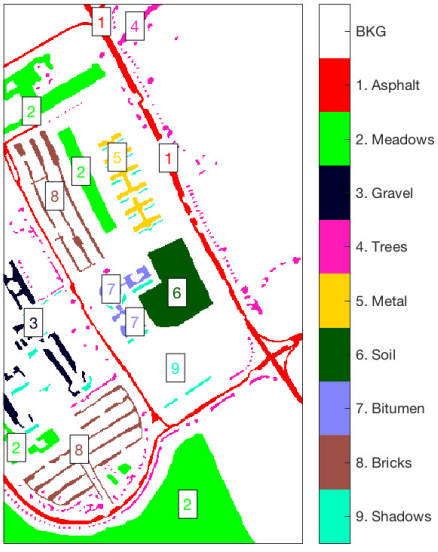


Fig. 4. University of Pavia along with ground truth image.

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TABLE I
INDIAN PINES TRAINING AND TEST SAMPLE COUNTS

Class Name	Training Samples	Test Samples	Class Name	Training Samples	Test Samples
Alfalfa	25	21	Soybean-notil	81	891
Corn-notil	83	1345	Soybean-mintil	99	2356
Corn-mintil	78	752	Soybean-clean	73	520
Corn	68	169	Wheat	70	135
Grass-pasture	79	404	Woods	90	1175
Grass-trees	78	652	Building	65	321
Grass-mowed	14	14	Towers	46	47
Hay-windrow	66	412	BKG	0	10,076
Oates	10	10			

TABLE II
SALINAS TRAINING AND TEST SAMPLE COUNTS

Class Name	Training Samples	Test Samples	Class Name	Training Samples	Test Samples
Broc. 1	67	1942	Corn	68	3210
Broc. 2	67	3659	Lettuce 4 Week	68	1000
Fallow	67	1909	Lettuce 5 Week	67	1860
Fallow Rough	69	1325	Lettuce 6 Week	67	849
Fallow Smooth	67	2611	Lettuce 7 Week	67	1003
Stubble	67	3892	Vineyard U	70	7198
Celery	68	3511	Vineyard VT	67	1740
Grapes	69	11202	BKG	0	56,975
Soil Vineyard	68	6135			

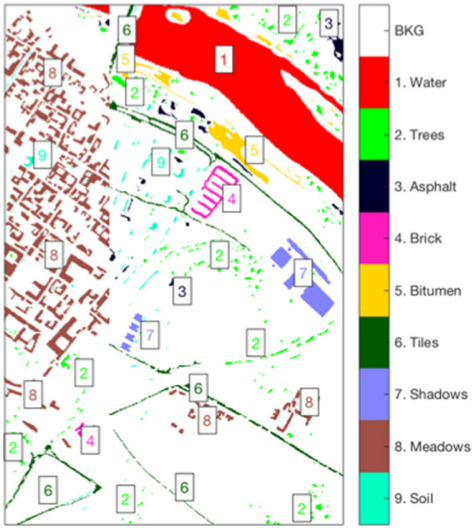


Fig. 5. University of Pavia ground truth image.

TABLE III
UNIVERSITY OF PAVIA AND TEST SAMPLE COUNTS

Class Name	Training Samples	Test Samples	Class Name	Training Samples	Test Samples
Asphalt	286	6345	Bare Soil	285	4744
Meadow	286	18363	Bitumen	285	1045
Gravel	285	1814	Bricks	285	3397
Trees	285	2779	Shadows	285	662
Metal	285	1060	BKG	0	164,624

TABLE IV
PAVIA CENTER TRAINING AND TEST SAMPLE COUNTS

Class Name	Training Samples	Test Samples	Class Name	Training Samples	Test Samples
Water	285	65686	Tile	285	8963
Trees	285	7313	Shadow	285	7002
Asphalt	285	2805	Meadow	285	42541
Brick	285	2400	Soil	285	2578
Bitumen	285	6299	BKG	0	635,488

10, to simulate the process of using larger step sizes. Finally, the root-mean-square (RMS) error of the OA was calculated between $\Delta_i = 1$ and all other step sizes $\Delta_i = 2$, 5, 7, and 10. A simple linear interpolation was used for the subsampled results to align them with the finely sampled results. It should be noted that for simplicity only a single experiment was conducted along with P_{OA} calculated on the training data. The step-size experimental results are shown in Fig. 6, and the RMS errors were calculated and displayed as legends. The results showed that the step size has minimal impact on the progression metric. The larger step sizes adequately tracked the performance progression and resulted in small

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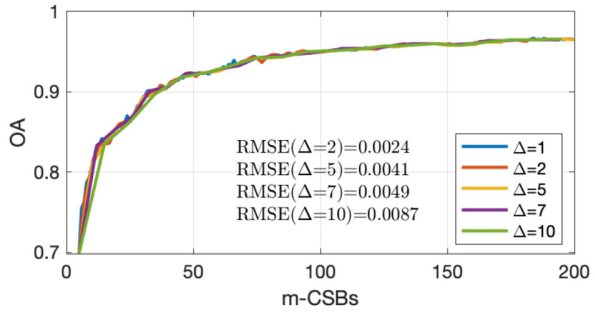


Fig. 6. Step size experiment for the Purdue Indian Pines image.

differences less than 1% of P_{OA} in RMS errors. These results suggested that the step size can be chosen completely based on the desired fidelity and the available computational resources.

C. Progressive Class Predictions

To fully illustrate the progressive classification sequence, a collection of classification maps and difference maps have been generated for each progression by diff(k, l) indicating the difference maps between classification maps produced by using k CBSs and l CBS. A single image from each sensor has been included in this section for illustration; however, the results were consistent for all four of the test images.

The progressive band classification sequence for the Indian Pines image is shown in Fig. 7 where the ground truth and the full-band predictions are also shown in the top left corner in Fig. 7 for comparison. The progressions from $m = 5$–30, $m = 3$–60, and $m = 65$–90 are grouped together in the top two rows, middle two rows, and bottom two rows, respectively. It is clearly shown from the difference maps that the class predictions rapidly changed in the early iterations, with large clusters of pixels changing within the first 20 CSBs. While the majority of the pixels eventually converged, there is still a small number of pixels that continued to change through 90 CSBs. Given the fact that the Purdue Indian Pines is more complicated than the other three test images, its classification accuracy is in the lower 90% range, it is not unreasonable to expect for some pixels to continually change, even for the larger number of CSBs. Nevertheless, the classification maps, with as few as 20 CSBs, provide a useful view of the scene and could be used to inform real-time decision making for many potential applications.

Fig. 8 shows the progressive band classification sequence for the University of Pavia. Similar to the Purdue Indian Pines, there multiple large clusters of pixels that changed within the first 20 CSBs; however, the class predictions began to converge around 30 CSBs and showed minimal changes beyond that point.

For both image scenes, the progressive classification sequence showed that there was a large amount of useful information available, even with as few as 20 CSBs. The current class predictions and knowledge of which data samples were changing between progressions can be coupled to inform many different real-time decisions. The difference maps could also be used *in situ* to isolate a subset of pixels for further analysis or transmission in a remote system. Additional experiments were also conducted for another AVIRIS data, Salinas in Fig. 3 and ROSIS Pavia Center in Fig. 5. Since similar conclusions can be also drawn from these experiments, their results are not included here.

D. Progressive Metric Correlation

The following experiments were conducted to quantify the effectiveness of CSBPCM as indicators of performance convergence. More specifically, how reliably could the progression metrics be used to determine when there are enough CSBs collected *in situ*. In this section, P_{OA} and P_{AA} were used to measure the performance of the progression metrics for a single image from each sensor. Nevertheless, the results were, indeed, consistent across all the four test images. The Pearson correlation coefficient, $\mu(a, b)$ defined as follows:

$$\mu(a, b) = \frac{\mathrm{cov}(a, b)}{\mathrm{std}(a)\mathrm{std}(b)}. \tag{16}$$

Equation (14) was used to quantify the utility of a progression metric for any two arbitrary vectors, a and b where std is the standard deviation. For each of the plots in Figs. 9–12, the solid blue and red lines represent P_{OA} and P_{AA}, respectively. The dashed lines represent the various progression metrics: P_{PGOA}(yellow), P_{PGAA} (purple), P_{PGAPR} (green), and P_{OTC} (cyan). Furthermore, the Pearson correlation coefficient was calculated for each combination of accuracy and progression metric, and is annotated directly on the plots.

Figs. 9 and 10 show a comparison between the classification accuracy and the progression metrics for the Purdue Indian Pines image without/with background pixels, respectively. For both cases, all of the progression metrics correlated very well with the classifier performance. There is a notable turning point in all of the curves around 25 CSBs from which a more gradual increase has been observed until it reached 100 CSBS in which the curves converged and the progression metrics resulted in a correlation coefficient greater than or equal to 0.94 and a maximum correlation of 0.99.

Similarly, Figs. 11 and 12 also show a comparison between the classification accuracy and the progression metrics for the University of Pavia image. This is another example of how background pixels had negative impact on the performance of the progression metrics. Interestingly, this is also an image that shows the largest discrepancies between the different progression metrics. When the background pixels were included, there was a difference of 0.22 between the lowest and the highest correlation coefficients.

There are a few notes worth mentioning with regard to the background pixels. First, excluding the background pixels in the progression metrics will result in a more optimistic estimate of performance, with a higher correlation between the classifier performances. Since many of the background pixels tend to be mixtures of the other classes, this can create some instability in the progression metric as the assigned label will tend to jump around among the background pixels. Second, it is important to recognize that some additional training techniques could potentially be employed to improve

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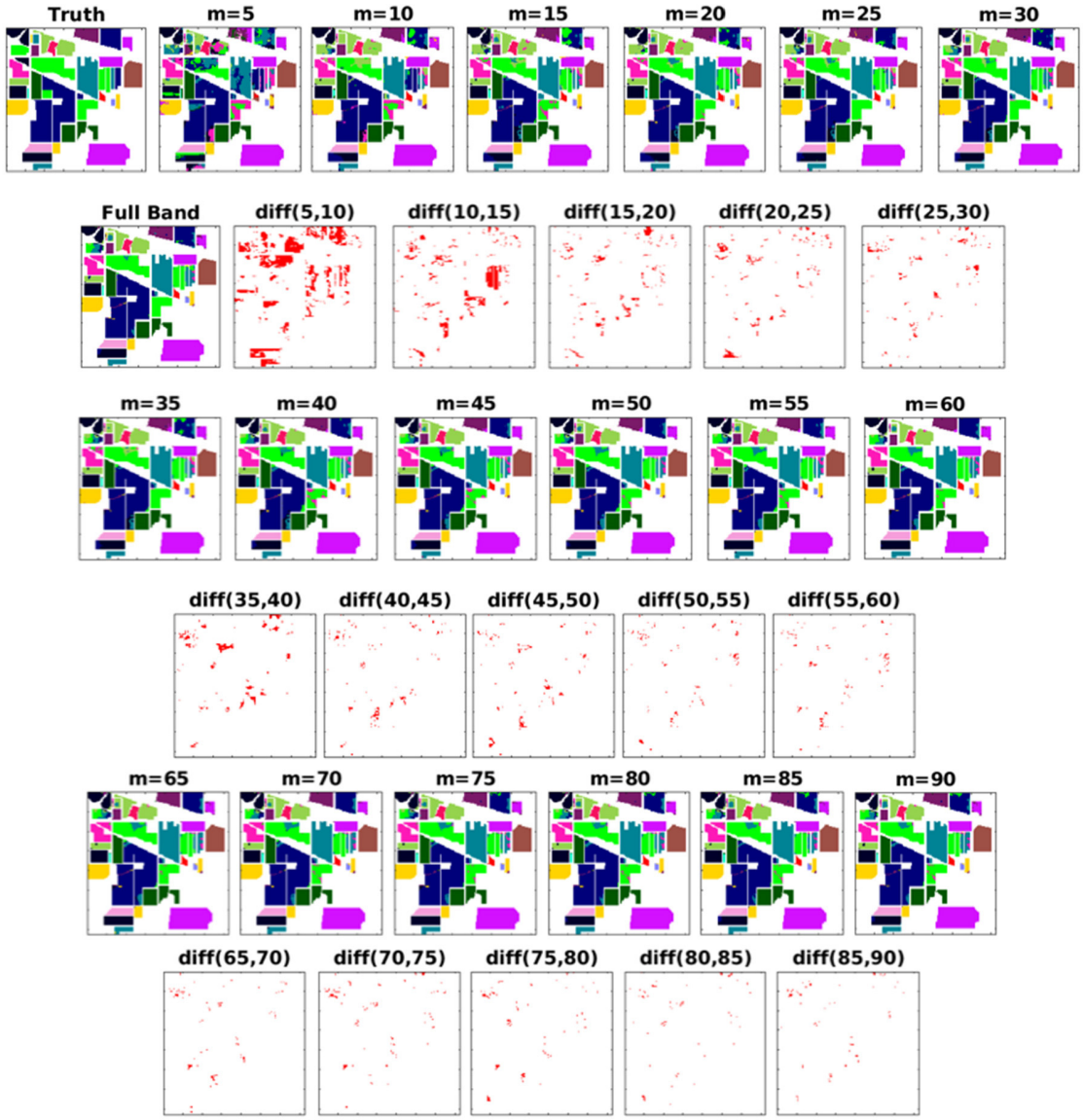


Fig. 7. Classification progression for the Purdue Indian Pines.

the classification robustness to the background pixels. This line of effort is left for future work.

E. Stopping Criteria

As final experiments, the proposed stopping rule was evaluated and applied to the progression metrics discussed in Section V-D to demonstrate the use of complete automation for the algorithm. The error threshold was set to a constant at $\varepsilon = 0.01$ and the progression metric threshold was varied from $\tau = 0.90$–099. The resulting CSBR, $P_{\mathrm{OA-EFF}}$ and $P_{\mathrm{AA-Eff}}$ are reported for each of the thresholds and displayed as histograms. For the cases where PCSBC failed to converge (i.e., stopping criteria were never satisfied), values were simply excluded from the plots. The following experiments were conducted both with and without the inclusion of background pixels.

Figs. 13–16 show the results for each of the images, without including background pixels, respectively. In general, the stopping criteria performed exceptionally well for all of the proposed progression metrics. For the Purdue Indian Pines, an efficacy of 0.95 was achieved with a CSBR less than 0.25. Full efficacy was achieved for the remaining three images with a CSBR less than 0.25 in all cases. The Salinas image showed the best performance with CSBR less than 0.2.

PCSBC also appears to be fairly robust to the specific choice of the progression metric threshold, τ. Increasing the threshold does not result in an overestimate of the number of CSBs. There was a single case, for the Indian Pines image, where PCSBC failed to converge with P_{ATC} and $\tau = 0.99$. This was due to the fact that $\mathrm{P_{ATC}}$ never reached a value of 0.99 (this is why no purple bar is shown for this case).

Regarding the choice of a particular progression metric, each of the proposed progression metrics provided fairly

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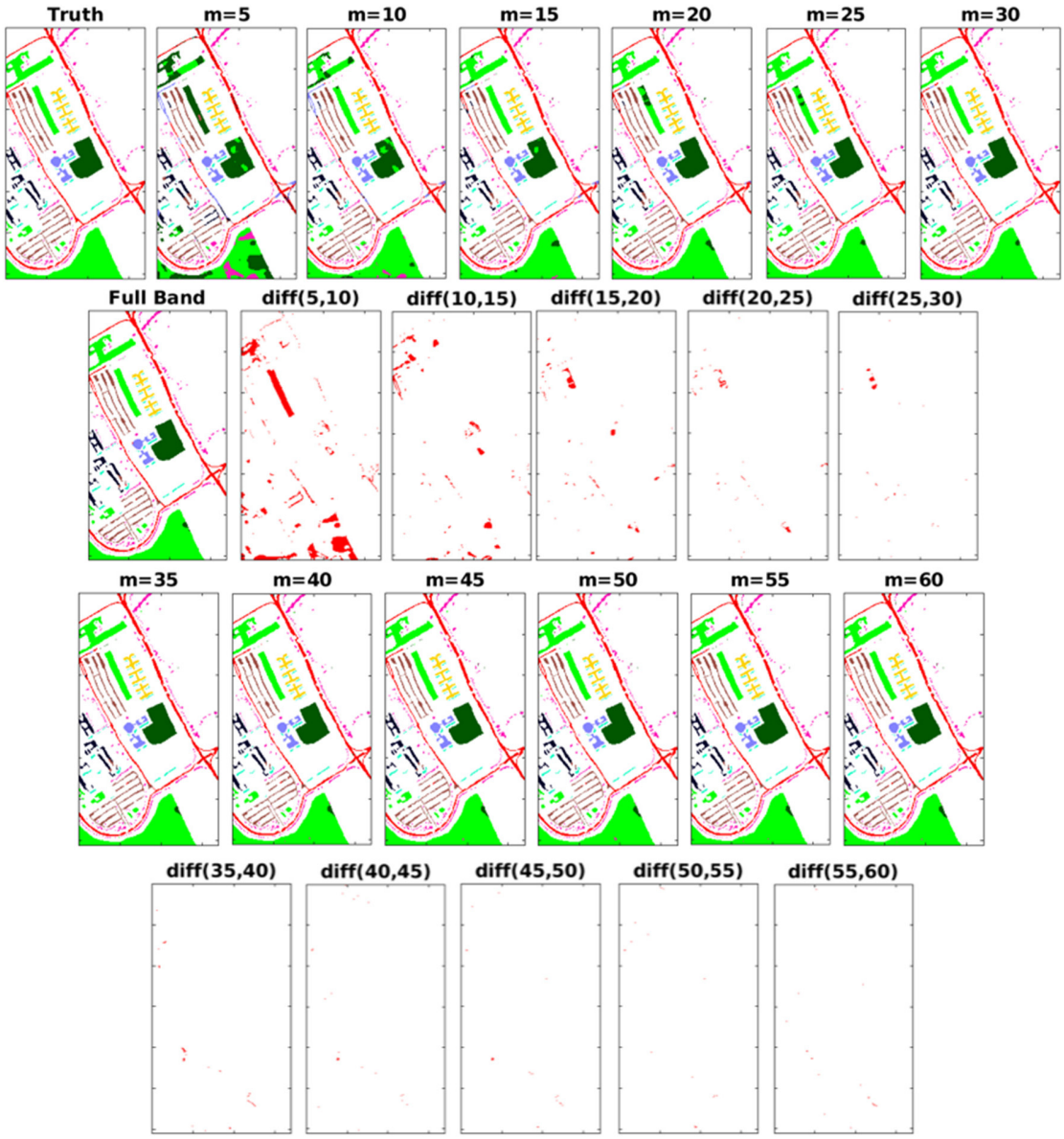


Fig. 8. Classification progression for University of Pavia.

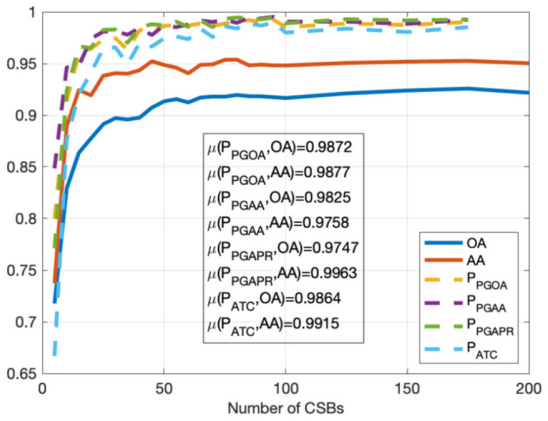


Fig. 9. Progression metrics for Indian Pines without background pixels.

similar results. There were no clear trends in Figs. 13–16 that would suggest one metric to be better than another. This is likely a value that would need to be tuned based on the sensor, the used classification algorithm and potentially evens the classes.

The experimental results with the inclusion of background pixels are shown in Figs. 17–20. When background pixels were included, there were two primary effects on the performance of PCSBC. First, the absolute value of the progression metrics also tended to decrease. This reduction directly affected the valid range of values that could be chosen for τ. To illustrate the effects on the threshold, we use the Purdue Indian Pines results in Fig. 17 for illustration where values of τ are displayed from 0.85 to 0.99. In this case, the lower threshold values resulted in CSBR and performance efficacies that were similar to those reported when excluding the background pixels. For higher thresholds, less than half of the cases converged. This is because the progression metrics never exceeded the thresholds. Interestingly, P_{PGOA} and P_{PGAA} converged in all cases but one. Conversely, P_{PGAPR} converged less than half of the time and P_{ATC} only converged once. The Indian

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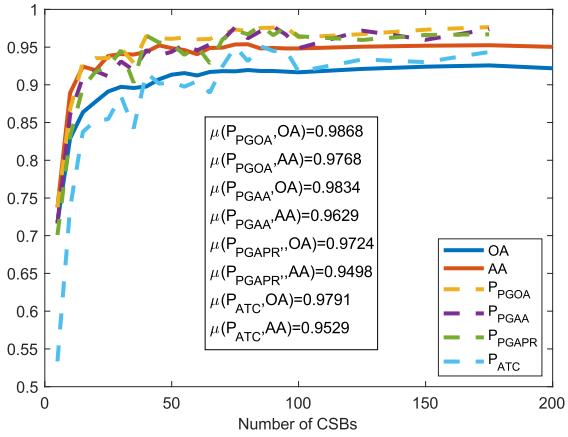


Fig. 10. Progression metrics for the Purdue Indian Pines with background pixels.

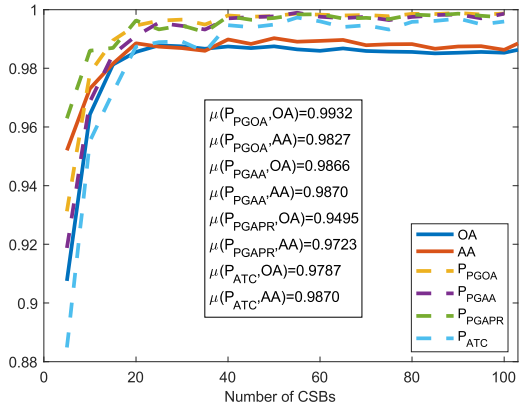


Fig. 11. Progression metrics for Pavia University without background pixels.

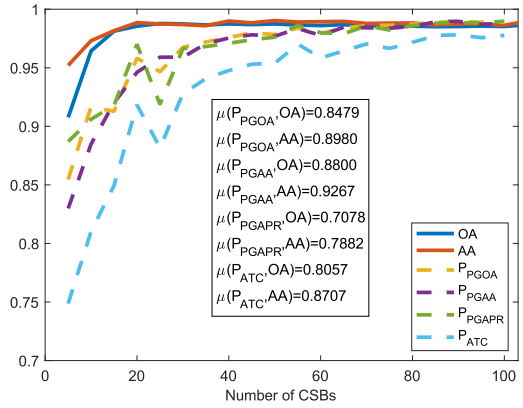


Fig. 12. Progression metrics for University of Pavia with background pixels.

Pines image was clearly the most challenging data among the four images that were used for experiments. For the other three images, PCSBC converged in all cases, except for the largest value of τ, 0.99. This can be once again attributed to the fact that many of the background pixels do not belong to a single class and tend to keep switching between classes iteration by iteration.

Second, as it was shown in Section V-D, the correlation between the progression metrics and classification performance decreases. For the University of Pavia, it introduced an obvious lag between the classification performance improvement and the progression metric. This lag would manifest as

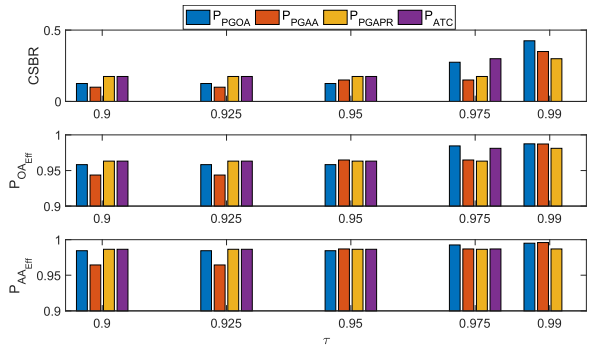


Fig. 13. CSBR and efficacy for Indian Pines without BKG pixels.

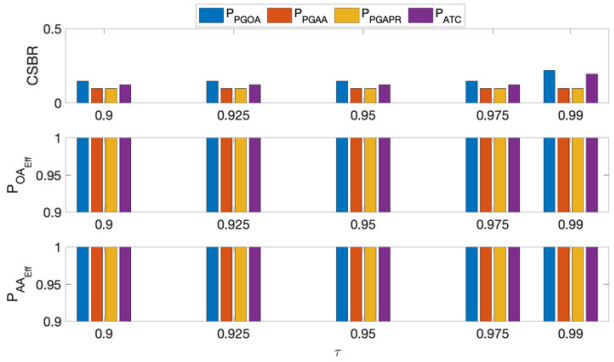


Fig. 14. CSBR and efficacy for Salinas without BKG pixels.

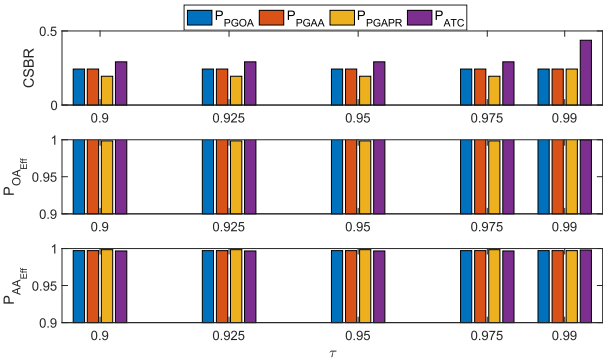


Fig. 15. CSBR and efficacy for Pavia University without BKG pixels.

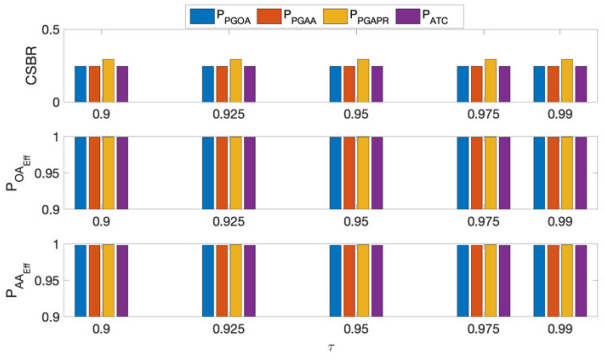


Fig. 16. CSBR and efficacy for Pavia Centre without BKG pixels.

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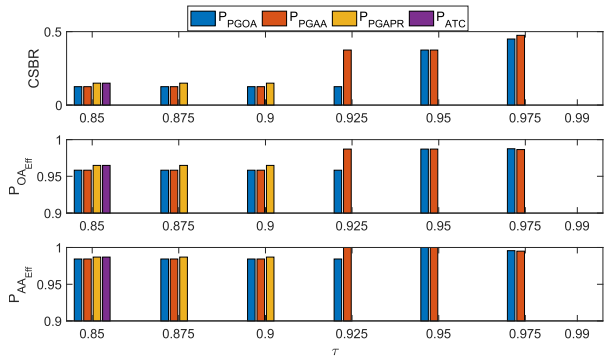


Fig. 17. CSBR and efficacy for Indian Pines with BKG pixels.

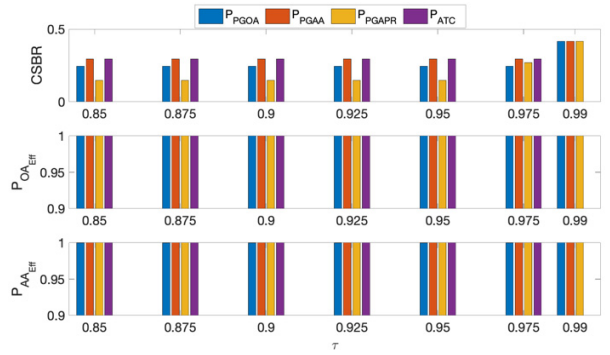


Fig. 18. CSBR and efficacy for Salinas with BKG pixels.

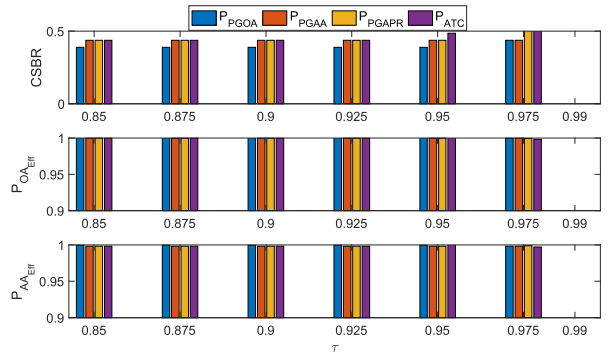


Fig. 19. CSBR and efficacy for University of Pavia with BKG pixels.

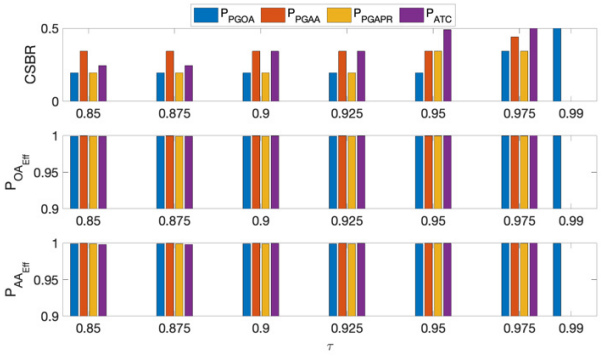


Fig. 20. CSBR and efficacy for Pavia Centre with BKG pixels.

an overestimate of the required number of CSBs and thus resulted in a higher CSBR. Such an effect is demonstrated in Fig. 19 with the University of Pavia. For the case where background pixels were excluded, a full efficacy can be achieved with CSBR around 0.25. However, with the inclusion of background pixels, it would require CSBR being nearly doubled to maintain full efficacy. Some smaller increases can also be noted for the other three images; however, they were not as severe as the University of Pavia image data. Nevertheless, for all the four image data sets, PCSBC was able to successfully converge to the desired efficacy with an appreciable reduction in the required number of CSBs.

V. Conclusion

This article presents an adaptive approach to estimating an appropriate number of CSBs for HSIC. It couples a band sequential processing with a progressive CS processing algorithm via PCSBC. It is this PCSBC which provides intermediate class predictions that could be used for decision making as well as adaptively determines when an enough number of CSBs have been collected. PCSBC can be implemented in three components, a progressive classifier, various progression metrics, and stopping criteria. Each of these components was discussed in detail and their corresponding experiments were also performed to analyze its performance.

Several important conclusions can be drawn from the experiments. First, the algorithm is not sensitive to the step-size between iterations. This allows the step-size to be chosen based on practical constraints, such as computational resources and system latency. Second, all of the progression metrics showed very high correlation with classification performance. The effects of background pixels were specifically considered and investigated. As shown by experiments for the Purdue Indian Pines and University of Pavia such an effect drastically reduced correlation. The reason for this decorrelation is that background pixels that fall near class boundaries tend to jump back and forth between classes throughout the progressions. Finally, the proposed stopping criteria were also demonstrated to be well suited for PCSBC. In general, the thresholds were fairly robust across all of the conducted experiments. For the case without background pixels, PCSBC was able to achieve nearly full efficacy for all four of the images while maintaining a CSBR less than 0.25. The case with inclusion of background pixels was also considered and some isolated effects were noted.

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